

Student Segmentation and Personalized Course Recommendation System for EduPro Using Machine Learning

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Abstract—Online learning platforms generate large volumes of learner data, making it challenging to provide personalized learning experiences using traditional recommendation methods. This project presents EduPro a machine learning-based student segmentation and personalized course recommendation system designed to improve learner engagement and course discovery. The system analyzes learner behavior using features such as age, course preferences, enrollment frequency, spending patterns, course ratings, and learning history. Data preprocessing techniques, including feature engineering, one-hot encoding, feature scaling, and Principal Component Analysis (PCA), were applied to prepare the dataset for clustering. The K-Means clustering algorithm was used to group learners into behavioral segments based on similar learning characteristics. These segments were then utilized to recommend relevant courses according to each learner's preferred category and course level. A Streamlit-based interactive dashboard was developed to visualize learner profiles, exploratory data analysis, cluster distributions, segment comparisons, business insights, and personalized course recommendations. The proposed system demonstrates how data-driven learner segmentation can support personalized education, improve user experience, and assist educational platforms in making informed decisions through meaningful analytics and intelligence recommendations.

Keywords – Machine Learning, K-Means Clustering, Student Segmentation, Personalized Course Recommendation, Educational Data Mining, Streamlit, Data Analytics.

I. INTRODUCTION

Online learning platforms have transformed the way learners access education by providing flexible, affordable, and self-placed learning opportunities. With thousands of courses available across multiple domains, learners often face challenges in identifying courses that match their interests, learning goals, and skill levels. Generic recommendation methods frequently suggest the same popular courses to all users, resulting in lower learner engagement, reduced course completion rates, and decreased satisfaction.

Personalized recommendation systems have become an important solution to address these challenges. By analysing learner behavior, preferences, and historical enrollment patterns, machine learning techniques can generate recommendations that are more relevant to individual learners. In addition, understanding the different types of learners through segmentation enables educational platforms to design targeted learning strategies and improve the overall learning experience.

Student segmentation is the process of grouping learning with similar behavioral and learning characteristics. Machine learning clustering algorithms, particularly K-Means clustering, are widely used for this purpose because they effectively identify hidden patterns within large datasets. By grouping learners based on factors such as age, spending behavior, course preferences, enrollment frequency, learning diversity, and average course ratings, educational platforms can better understand learner needs and provide personalized recommendations.

This research presents the development of EduPro, a machine learning-based student segmentation and personalized course recommendation system. The system utilizes learner transaction data, demographic information, and course details to perform extensive data preprocessing and feature engineering. Principal Component Analysis (PCA) is employed for dimensionality reduction, followed by K-Means clustering to identify distinct learner segments. Based on the assigned learner segment and individual preferences, the system recommends suitable courses that align with the learner's interests and learning level.

To enhance usability, an interactive dashboard was developed using the Streamlit framework. The dashboard enables visualization of learner profiles, exploratory data analysis, cluster distributions, personalized course recommendations, comparative analysis of learner segments, and business insights. This provides both learners and educational administrations with valuable information for decision making.

The proposed system demonstrates how data-driven educational analytics can improve learner engagement, personalize learning experiences, and support better course discovery. By integrating clustering techniques with recommendations strategies, the project provides a scalable solution that can be adapted to modern online learning platforms.

The remainder of this paper is organized as follows. Section II presents the literature review of existing student segmentation and recommendation systems. Section III describes the proposed methodology. Section IV explains the implementation details. Section V discusses the experimental results and analysis. Finally, Section VI concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

A. Introduction and Significance

The rapid growth of online learning has produced a vast catalog of courses, making it increasingly difficult for students to choose the most suitable ones. As Narimani and Barberà (2024) note, "as education has evolved towards online learning... the need to offer personalized learning experiences and content has never been greater". In response, course recommender systems have been developed to help learners "select courses that suit their interests and aptitude". Meanwhile, learner segmentation (clustering students by behavioral patterns) enables tailored support for different learner groups (e.g. at-risk vs. engaged) instead of a one-size-fits-all approach. Combining segmentation with recommendation promises more targeted personalization. For example, the EduPro initiative cited above envisions a unified ML pipeline that "identifies groups of learners who share behavioral patterns" and then "suggest[s] what they should study next". In short, the literature indicates that clustering (to profile learners) and personalized recommendation are both key strategies for improving engagement and outcomes in e-learning. The following review examines prior work on each of these aspects - student segmentation and course recommendation - and on methods that integrate them, to identify strengths, limitations, patterns and gaps related to our objective of segmenting learners by behavior and giving them personalized course suggestions.

B. Learner Segmentation and Behavioral Profiling

Many studies use unsupervised learning (especially K-means clustering) to segment students by behavior or performance. For instance, Ye et al. (2023) applied K-means to LMS activity data from 127 students and successfully clustered them into distinct "high", "moderate" and "low" engagement profiles. This work highlights how clustering real-time engagement metrics (e.g. quiz attempts, forum posts, time-on-task) can automatically identify at-risk learners or periods of disengagement, enabling timely interventions. Similarly, Adithya (2026) used K-means with PCA to partition synthetic student data into six behavioral clusters. Among eleven classifiers for course difficulty prediction, Random Forest achieved 98.85% accuracy in this setup – albeit on engineered synthetic data – and the K-means solution had moderate separation (silhouette ≈ 0.50). Adithya's work demonstrates that

segmentation can uncover meaningful learner groups, but also warns that results may not generalize beyond simulated datasets.

Others have emphasized segmentation as foundational to personalization. Baker and Siemens (2013) and subsequent reviews argue that clustering enables differentiated support for learner subgroups. In fact, Mhagama and Garg (2025) systematically surveyed educational recommender systems and found that clustering (especially K-means) is one of the most widely adopted techniques in educational data mining, alongside ensemble classifiers. They report that ML approaches (including clustering) dominate over pure collaborative filtering in published systems. This trend is mirrored in empirical studies: for example, clustering students by engagement or background features is common practice in MOOCs and blended courses. However, most segmentation work to date focuses on static or coarse features (course usage counts, grades, demographics). Narimani and Barberà (2024) point out a key limitation: few educational studies incorporate learners' micro-behaviors (fine-grained actions) into profiles, a gap we aim to address. In summary, the literature shows that learner clustering is a standard tool, but often with limited features and without linking clusters directly to downstream recommendations.

C. Personalized Course Recommendation Systems

Another strand of research develops systems that recommend courses based on student profiles and course features. Classic approaches include content-based and collaborative-filtering recommenders (using ratings, grades or metadata). For example, Kalokhe and Kumbhar (2024) designed an "add-on" course recommender by analyzing a survey of 302 undergraduates. They used K-means to segment students by preferences and interests, and descriptive statistics to guide which supplementary courses to recommend (finding a strong interest in practical fields like Data Science and Cybersecurity). Their work demonstrates learner profiling for recommendations, but is essentially survey-based and descriptive: it identifies popular student interests (94.7% wanted extra courses) but does not evaluate algorithmic performance on actual recommendation accuracy.

More sophisticated methods have also been proposed. Asadi, Jafari and Shokrollahi (2019) explicitly combined clustering with association-rule mining to build a course recommender. They first clustered students into two groups by background/skills, then applied fuzzy association rules on each cluster's course enrollment data. This approach produced "appropriate recommendations and a predicted score" for each student. While their dataset (2004–2015 records from a Tehran college) is limited and results are not benchmarked by precision/recall, the study showed that clustering can inform rule-based recommendations tailored to student profiles. Similarly, Rahman et al. (2022) used K-means to identify the most- and least-popular courses among undergraduates and then applied the FP-Growth algorithm to generate course suggestions. Their "add-on course" recommender outperformed simple baselines, indicating that embedding

popularity patterns via clustering+FP-Growth can yield relevant recommendations, although again quantitative metrics were sparse in the publication.

In general, course recommendation research has increasingly embraced machine learning and hybrid methods. Mhagama & Garg (2025) found that besides clustering, collaborative filtering (CF) and hybrid CF/content methods are also common. However, collaborative approaches suffer from cold-start issues (new students or courses), so researchers have tried content-based and mixed approaches. For instance, recent MOOC recommenders augment CF with course attributes or knowledge models. One example is a hybrid MOOC recommender by Wen and Li (2023) that first clusters learners and then applies user-based CF within each cluster. On Coursera data, their Cluster-CF method achieved higher F1 scores (≈ 7 – 17% relative gain) and much faster runtime than a non-clustered CF. This suggests that pre-clustering users not only reduces data sparsity but also improves recommendation effectiveness. Algami and Sheldon (2025) took yet another tack: they built a multi-modal pipeline that predicts student dropout and also ranks entire degree programs (using RNNs and BERT) for new students. Their system reached high top-10 accuracy ($\text{NDCG}@10 \approx 0.58$, $\text{Hit}@10 \approx 0.94$) without any prior interaction data, highlighting how advanced ML (and transfer learning) can power cold-start course recommendations. These works indicate strong progress in ML-driven recommenders, but also reveal varied scopes (courses vs. programs, user-side vs. content-side) and benchmarks, making direct comparisons difficult.

D. Integrated Segmentation-Recommendation Approaches

Crucially for our goals, a growing subset of work explicitly integrates learner segmentation with the recommendation engine. Asadi et al. (2019) and Rahman et al. (2022) (above) fall into this category by feeding clusters into rule-based recommenders. More recently, Adithya's EduPro system (2026) exemplifies a full pipeline: after K-means clustering (six groups), it conditions course-difficulty prediction and recommendations on cluster membership, and even provides an interactive dashboard. Although preliminary on synthetic data, this pipeline aligns strongly with our objectives – it shows how segmentation and recommendation can be combined in one framework. Similarly, Wen and Li (2023) showed empirically that cluster-conditioned recommendation outperforms unclustered baselines. Their MOOC study is particularly relevant: learners were divided into clusters based on course preferences and capabilities, then user-CF was applied within clusters. This Cluster-CF hybrid not only improved precision/recall over plain CF but also cut runtime by $\sim 50\%$. These results reinforce the intuition that clustering students first (segmentation) can concentrate similar behaviors and yield more accurate, efficient recommendations than treating all users homogeneously.

Other frameworks from the broader recommender literature also support this idea. For example, Ghiasabadi et al. (2022)

built a movie recommender where items were clustered and a learning-automata model updated user interest in each item-cluster. They reported that this adaptive, clustered approach significantly outperformed standard methods on precision and recall. While in a different domain, it underscores how combining unsupervised grouping (users or items) with a recommender can boost performance. In contrast, many traditional educational recommenders rely solely on collaborative or content filtering without an explicit clustering stage, which can be a drawback in sparse data regimes.

E. Synthesis, Patterns and Gaps

Across the surveyed studies, clear patterns emerge. First, machine learning and clustering are pervasive in recent educational recommender research. Many systems employ K-means or similar algorithms for segmentation, often as a precursor or supplement to a recommendation algorithm. Content-based and collaborative approaches remain popular but are frequently hybridized with ML models. Second, studies that combine segmentation with recommendation generally report improved performance (Rahman 2022; Wen & Li 2023) or usability (Adithya 2026), suggesting the value of our integrated approach. Third, ensemble classifiers (e.g. Random Forest) tend to excel at predicting educational outcomes (as in Adithya's 98.85% result), and are often part of recommenders' pipelines, though these high accuracies are sometimes due to synthetic or highly processed features.

Contradictions and limitations are also evident. Many approaches achieve high accuracy on curated datasets but may not generalize well. For example, Adithya (2026) cautions that his 98.85% classification accuracy is largely due to the “structured nature of the synthetic data”. Similarly, Algami & Sheldon's (2025) impressive ranking metrics apply to a specific university program dataset. Conversely, simpler methods (e.g. Kalokhe & Kumbhar's survey analysis) yield little quantitative performance data. There is also diversity in the target of “courses” – some recommend supplemental courses (Kalokhe 2024), others entire degree programs (Algami 2025) or general MOOCs (Wen & Li 2023) – making it hard to directly compare results. Moreover, despite frequent use of clustering, few studies deeply analyze cluster quality or interpretability. Adithya (2026) reports only a moderate silhouette score (~ 0.50), and many papers do not report clustering metrics at all. In terms of learner modeling, most systems still rely on aggregate features (grades, ratings) or self-reported preferences; as Narimani & Barberà (2024) note, fine-grained behavioral data (“micro-behaviors”) are underexplored. Finally, existing literature often targets traditional students; as Mhagama & Garg (2025) emphasize, very few recommenders address educators, administrators, or non-traditional (lifelong/workplace) learners.

In sum, the literature demonstrates significant progress toward ML-driven personalization in education, but also reveals gaps. Many systems apply clustering and recommendation

separately, and only a few integrate them end-to-end with empirical evaluation. There is a need for approaches that fuse rich behavioral profiling with recommendation logic, validated on real platform data. Our proposed research seeks to fill this gap. We will engineer multi-dimensional features capturing learners’ content usage and engagement (addressing the “micro-behaviors” gap), apply clustering to segment learners by actual behavior, and then generate personalized course suggestions for each segment. By combining segmentation and recommendation in one pipeline – with thorough metric evaluation and visualization (as in Adithya 2026) – we aim to create a more comprehensive, interpretable system. This aligns with the identified needs for context-aware, user-centered recommenders, and directly addresses the objective of enhancing engagement through tailored course paths.

In conclusion, previous studies provide valuable techniques and insights (e.g. clustering, ensemble ML, hybrid recommenders) but remain fragmented. Our review shows that while clustering-based personalization can improve recommendation quality, few works fully leverage both behavioral segmentation and modern ML in a unified way. By critically synthesizing these studies, we see that our research – building on clustering plus recommendation to boost engagement – is both novel and necessary. It aims to correspond closely to the stated objectives where prior literature has only provided partial solutions. Specifically, we target the noted shortcomings (synthetic data, lack of micro-behaviors, limited validation) and thus will contribute an end-to-end, data-driven recommender that bridges the current gaps highlighted by Narimani et al. (2024) and Mhagama et al. (2025).

III. METHODOLOGY

A. System Overview

The proposed EduPro Student Segmentation and Personalized Course Recommendation System was developed to analyze learner behavior and provide personalized course recommendations using machine learning techniques. The system consists of six major stages: dataset collection, data preprocessing, feature engineering, dimensionality reduction, learner segmentation, and personalized course recommendation. Finally, an interactive dashboard was developed using Streamlit to visualize learner analytics, clustering results, and personalized recommendations. Figure 2 illustrates the overall workflow of the proposed methodology.

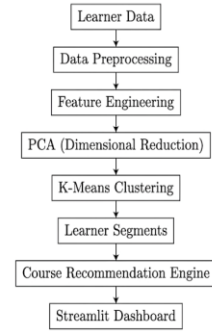


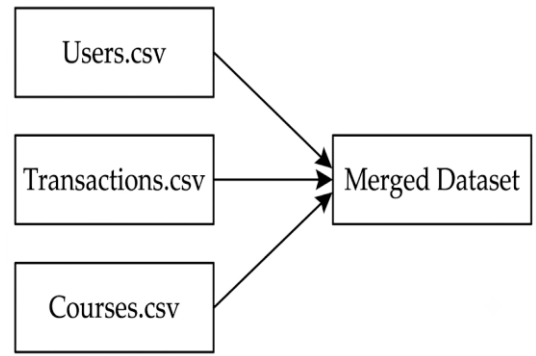
Figure 1: Machine learning and deployment workflow for personalized course recommendation.

B. Dataset Description

The dataset used in this study was created by integrating learner, course, and transaction information collected from the EduPro online learning platform. Three CSV files were merged using common identifiers to construct a comprehensive learner dataset suitable for behavioral analysis.

The Users dataset contains **3,001 learner records** with demographic information such as user identification, age, gender, and email. The Transactions dataset consists of **1,001 learner transactions**, including enrollment information, payment details, transaction amount, and enrollment dates. The Courses dataset contains **61 course records**, describing course names, categories, levels, durations, and ratings.

The merged dataset combines demographic characteristics, learning preferences, enrollment history, financial information, and course attributes into a single dataset that supports clustering and personalized recommendation.



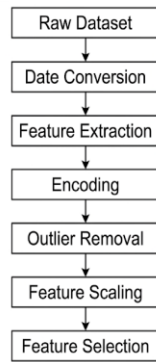
C. Data Preprocessing

Data preprocessing was performed to improve the quality of the dataset before applying machine learning algorithms. Since the datasets contained no duplicate records or missing values, no data imputation or duplicate removal was required. However, several preprocessing techniques were applied to transform the raw data into a machine-learning-ready format.

The TransactionDate attribute was converted into datetime format, enabling the extraction of additional temporal features

such as Year, Month, Day, and DayOfWeek. These temporal attributes provide useful information regarding learner enrollment behavior over time.

Categorical attributes including gender, payment method, course category, course level, and preferred learning preferences were converted into numerical representations using Label Encoding and One-Hot Encoding. Numerical features were normalized using StandardScaler to ensure that variables with different scales contributed equally during clustering. Furthermore, the Interquartile Range (IQR) method was applied to reduce the influence of extreme values, thereby improving clustering performance. Finally, feature selection was performed to retain only the most relevant attributes required for learner segmentation.



D. Feature Engineering

Feature engineering was performed to generate meaningful behavioral indicators that better represent learner characteristics than the original attributes alone. These engineered features provide valuable information regarding learner engagement, learning preferences, spending behavior, and course diversity.

The generated features include:

The generated features include Total_courses_enrolled, AvgCoursesPerCategory, Enrollment_frequency, PreferredCategory, PreferredLevel, AvgCourseRating, AvgSpending, DiversityScore, LevelScore, LearningDepthLevel, Age_group, HighlyRated, ExcellentCourse, LongDurationCourse, and ShortDurationCourse.

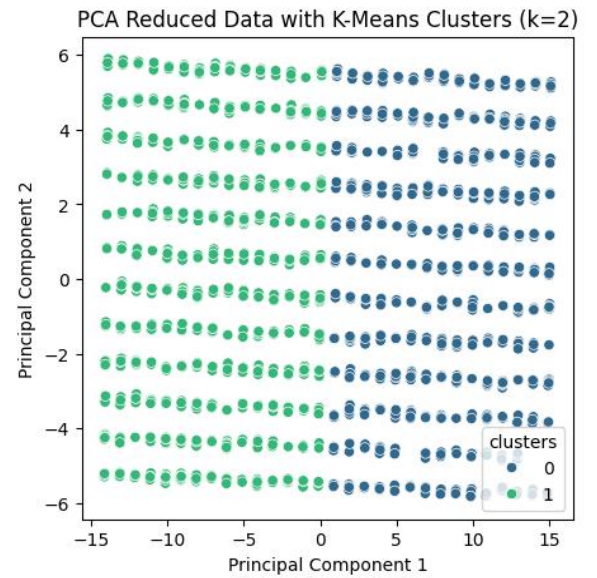
Temporal features such as Year, Month, Day, and DayOfWeek were also extracted from transaction dates to capture learner enrollment trends. These engineered variables significantly improved the representation of learner behavior, enabling more meaningful clustering results.

E. Dimensionality Reduction using Principal Component Analysis (PCA)

The processed dataset contained a large number of numerical features generated after feature engineering and one-hot encoding. To reduce data dimensionality while preserving the majority of the information, Principal Component Analysis (PCA) was applied.

PCA transformed the high-dimensional feature space into **two principal components**, which retained the most significant variance within the dataset. Reducing the dimensionality simplified visualization, decreased computational complexity, and improved the efficiency of clustering algorithms.

The transformed two-dimensional feature space was subsequently used for learner segmentation and visualization of cluster distributions.



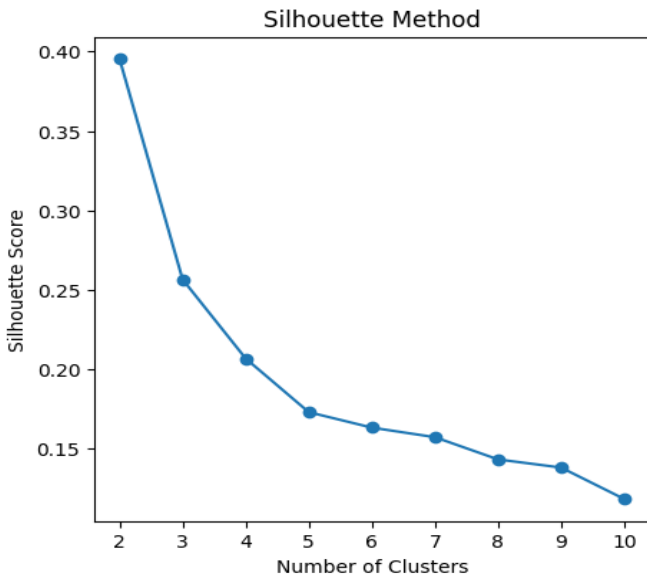
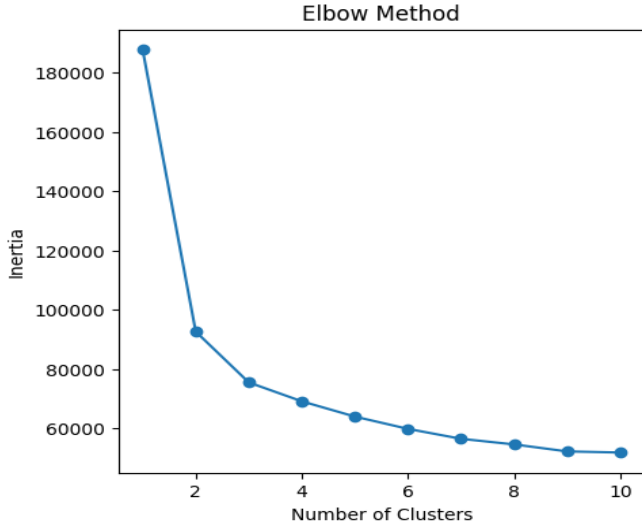
F. Learner Segmentation using K-Means Clustering

Learner segmentation was performed using the K-Means clustering algorithm. K-Means groups learners based on behavioral similarity by minimizing the distance between data points and their corresponding cluster centroids.

The optimal number of clusters was determined using the Elbow Method, where the Within Cluster Sum of Squares (WCSS) was calculated for different cluster values. The elbow point was observed at **K = 2**, indicating the most suitable number of learner segments.

The clustering quality was further evaluated using the Silhouette Score. The highest Silhouette Score (**0.3954**) was achieved for **two clusters**, confirming that this cluster configuration provided the best separation among learners. As the number of clusters increased, the Silhouette Score gradually decreased, indicating poorer cluster quality.

In addition to K-Means clustering, Hierarchical Clustering was performed to compare learner grouping patterns and validate the segmentation results.

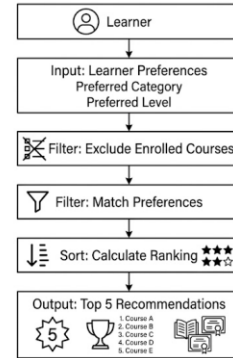


G. Personalized Course Recommendation

Following learner segmentation, a personalized recommendation engine was developed to suggest relevant courses based on learner preferences and behavioral characteristics. The recommendation process first identifies the learner's preferred course category and preferred learning level.

Courses that have already been enrolled by the learner are excluded from the recommendation list to prevent duplicate suggestions. The remaining courses are filtered according to preferred category and course level. Candidate courses are then ranked using course ratings, and the **Top 5 highest-rated courses** are recommended to each learner.

This recommendation strategy combines learner preferences with course quality, ensuring that suggested courses are both relevant and highly rated.



H. Streamlit Dashboard Development

An interactive web application was developed using the Streamlit framework to visualize learner analytics and recommendation results. The dashboard provides an intuitive interface for exploring learner behavior and cluster information.

The dashboard consists of eight modules:

- Home Dashboard
- Dataset Explorer
- Exploratory Data Analysis (EDA)
- Learner Profile Explorer
- Cluster Visualization Dashboard
- Personalized Course Recommendation
- Segment Comparison
- Business Insights

Interactive visualizations were created using Plotly, Matplotlib, and Seaborn to display learner distributions, course statistics, clustering results, and recommendation outcomes.

I. Methodology Summary

The proposed methodology integrates data preprocessing, feature engineering, dimensionality reduction, machine learning clustering, personalized recommendation, and interactive visualization into a unified framework. By combining K-Means clustering with learner preference analysis, the system effectively identifies learners with similar behavioral patterns and recommends suitable courses tailored to their interests and learning levels. The developed Streamlit dashboard further enhances usability by providing interactive analytics and supporting informed decision-making for educational institutions.

IV. RESULTS AND DISCUSSION

The proposed EduPro Student Segmentation and Personalized Course Recommendation System was successfully implemented using Python and machine learning techniques. The system integrated learner information, transaction records, and course details to identify learner behavior and generate personalized course recommendations. The developed Streamlit dashboard provided an interactive interface for visualizing learner profiles, clustering results, and recommendation outputs.

A. Data Preprocessing Results

The three datasets were successfully merged into a single dataset containing learner demographics, transaction history, and course information. During preprocessing, the transaction date was converted into a datetime format, enabling the extraction of useful temporal features such as year, month, day, and day of the week. Since the dataset contained no missing values or duplicate records, no imputation or duplicate removal was required.

Feature engineering generated several behavioral attributes, including total courses enrolled, average spending, preferred course category, preferred course level, enrollment frequency, diversity score, and learning depth level. These engineered features provided a better representation of learner behavior than the original dataset alone.

B. Clustering Performance

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset while preserving the most significant information. The transformed dataset with two principal components enabled easier visualization of learner distribution and improved the efficiency of the clustering algorithm.

The Elbow Method was used to determine the optimal number of clusters. The Within Cluster Sum of Squares (WCSS) decreased as the number of clusters increased, with a noticeable bend occurring at $K = 2$, indicating that two clusters provided the most suitable segmentation for the dataset.

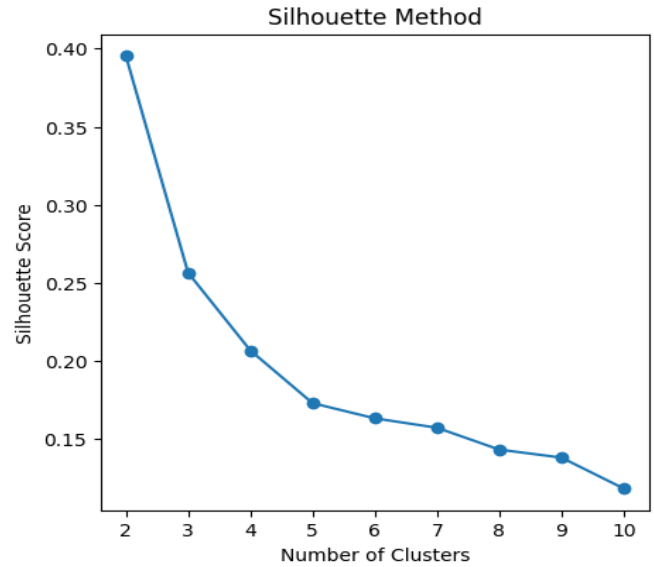
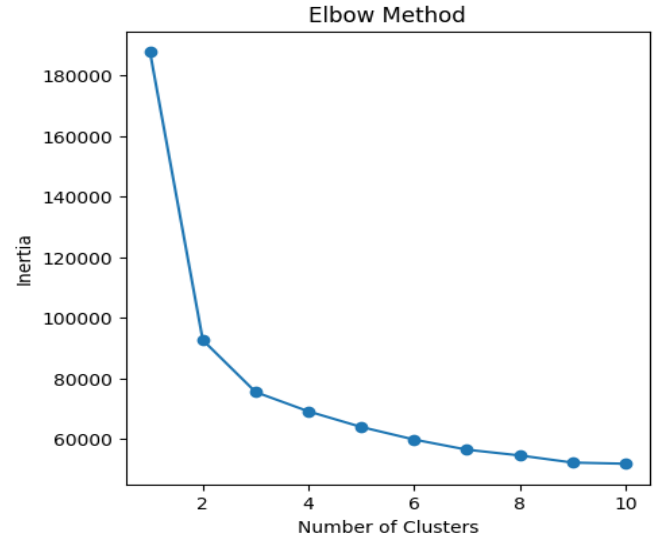
The clustering performance was further evaluated using the Silhouette Score. The results are presented in Table I.

Table I. Silhouette Scores for Different Numbers of Clusters

Number of Clusters	Silhouette Score
2	0.3954
3	0.2565
4	0.2067
5	0.1729
6	0.1632

7	0.1571
8	0.1431
9	0.1380
10	0.1182

The highest Silhouette Score was obtained when the number of clusters was set to two, indicating better separation between learner groups compared to larger cluster values. Based on both the Elbow Method and Silhouette Score analysis, two clusters were selected for learner segmentation.



C. Learners Segmentation Analysis

The clustering process grouped learners into two distinct behavioral segments based on their learning preferences and engagement patterns. Although both groups consisted of active learners, noticeable differences were observed in their enrollment frequency, spending behavior, and preferred learning categories.

One cluster primarily contained learners with higher engagement levels, characterized by frequent enrollments and greater average spending on courses. The other cluster consisted of learners who enrolled less frequently and generally focused on fewer course categories. These behavioral differences demonstrate that learner segmentation can effectively identify groups with similar learning characteristics, making it easier to provide targeted recommendations.

The visualization of clusters using PCA showed a reasonable separation between the learner groups, supporting the effectiveness of the clustering process.

D. Personalized Course Recommendation Results

The recommendation module generated personalized course suggestions by considering each learner's preferred course category, preferred learning level, previously enrolled courses, and course ratings. Courses that learners had already completed were excluded to avoid duplicate recommendations.

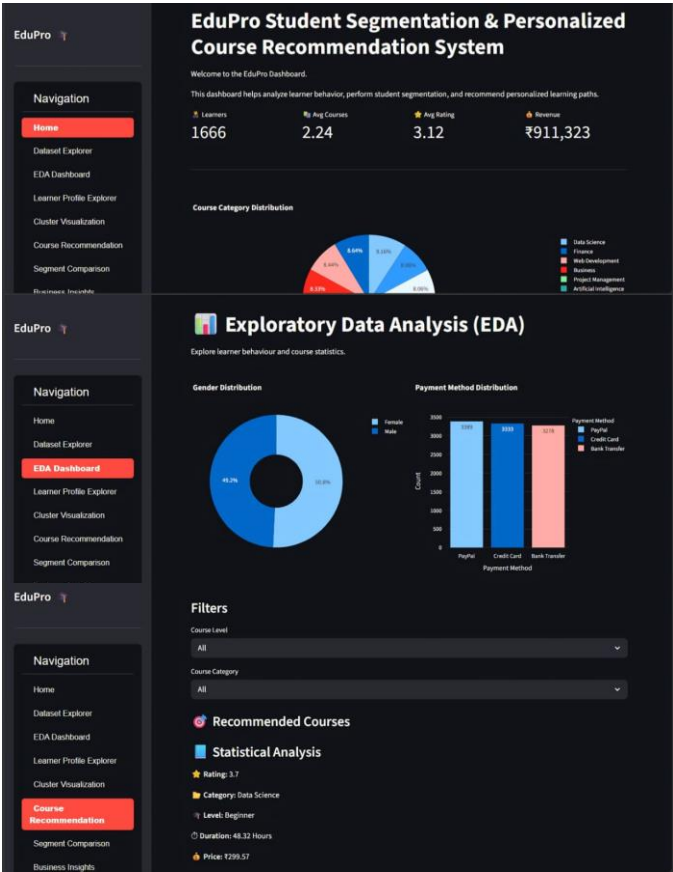
For each learner, the system recommended the top five highest-rated courses that matched their interests and learning preferences. Users could further filter recommendations based on course category and course level through the Streamlit dashboard.

Testing the recommendation module with different learner profiles showed that the recommended courses changed according to individual preferences, indicating that the recommendation process was personalized rather than identical for every user.

E. Dashboard Implementation

The developed Streamlit dashboard provided an interactive platform for exploring learner analytics and recommendation results. The application consisted of eight functional modules: Home, Dataset Explorer, Exploratory Data Analysis, Learner Profile Explorer, Cluster Visualization, Personalized Recommendation, Segment Comparison, and Business Insights.

The dashboard enabled users to analyze learner distributions, compare cluster characteristics, visualize behavioral trends, and obtain personalized course recommendations through an easy-to-use interface. Interactive charts created using Plotly, Matplotlib, and Seaborn improved the interpretability of the analytical results.



F. Discussion

The results demonstrate that integrating feature engineering with machine learning techniques can effectively support learner segmentation and personalized recommendation. The engineered behavioral features contributed to more meaningful learner profiles by capturing patterns beyond basic demographic information. The combination of PCA and K-Means clustering simplified the clustering process while maintaining meaningful group separation.

Unlike traditional recommendation approaches that rely mainly on popular courses, the proposed system considers individual learner behavior and preferences when generating recommendations. This allows learners to receive suggestions that better align with their interests and learning goals.

Although the proposed system achieved effective learner segmentation, several limitations remain. The dataset used in this study was relatively limited in size and contained only selected learner attributes. Additional features such as course completion rates, assessment scores, learner feedback, and time spent on courses could further improve the quality of segmentation and recommendations. Future work may also investigate advanced recommendation techniques, including collaborative filtering, hybrid recommendation models, and

deep learning approaches, to further enhance recommendation accuracy and personalization.

V. CONCLUSION

This study presented the design and implementation of the **EduPro Student Segmentation and Personalized Course Recommendation System** using machine learning techniques. The proposed system successfully integrated learner demographic information, transaction history, and course details to analyze learning behavior and provide personalized course recommendations. Through comprehensive data preprocessing, feature engineering, Principal Component Analysis (PCA), and K-Means clustering, learners were grouped into meaningful behavioral segments. These segments formed the basis for recommending relevant courses that matched each learner's preferences, learning level, and previous enrollment patterns.

The implementation of an interactive Streamlit dashboard further enhanced the usability of the system by allowing users to explore learner profiles, visualize clustering results, compare learner segments, and access personalized recommendations through an intuitive interface. The use of the Elbow Method and Silhouette Score supported the selection of an appropriate number of clusters, contributing to meaningful learner segmentation.

Overall, the developed system demonstrates how machine learning can support personalized learning experiences and assist educational platforms in understanding learner behavior more effectively. By combining data analytics with recommendation techniques, the project provides a practical approach to improving learner engagement and supporting informed decision-making in online education.

Although the system achieved its intended objectives, there is scope for further improvement. Future work may incorporate larger and more diverse datasets, additional learner behavioral attributes, and advanced recommendation techniques such as collaborative filtering, hybrid recommendation models, or deep learning approaches. Integrating real-time learner interactions and feedback could also enhance the accuracy and adaptability of future recommendation systems.

VI. FUTURE WORK

Future enhancements to the proposed system may include the integration of real-time learner activity, course completion rates, assessment performance, and learner feedback to improve recommendation quality. Advanced recommendation techniques, including collaborative filtering, hybrid recommender systems, and deep learning models, can also be explored to increase personalization. In addition, deploying the application on a cloud platform and integrating it with a Learning Management System (LMS) would improve

scalability and enable practical use in real-world educational environments.

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